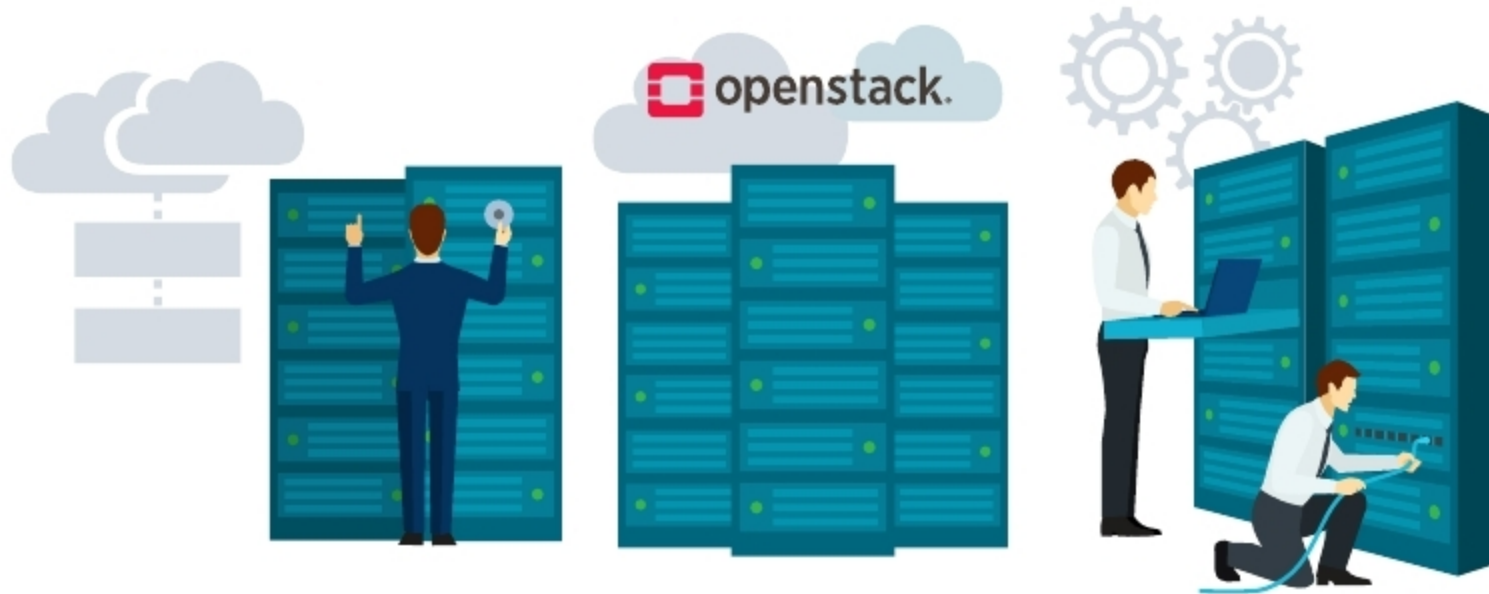


Top Open Source Tools to Help You Manage OpenStack Servers

OpenStack is an open source platform comprising a set of software tools to build and manage cloud computing platforms for public and private cloud infrastructure. Managing OpenStack's functionality at the enterprise level is a prime requirement for today's large scale cloud deployment. Systems administrators have a hard time when selecting open source tools for the management of OpenStack.



In recent times, software production companies require continuous deployments and often operate over distributed systems scattered across the world in a random manner. In the world of the cloud, the basic and prime fundamental is to automate everything when it comes to on-demand cloud services. Various commercial companies like Red Hat and Rackspace provide automation services, but OpenStack cloud technology is also a significant option.

OpenStack is an open source platform comprising a set of software tools for building and managing cloud computing platforms for public and private cloud infrastructure. The software platform comprises interrelated components that control diverse, multi-vendor hardware pools of processing, storage and networking resources throughout a data centre.

The primary advantage of OpenStack is that it is designed for horizontal scalability so it becomes easy to add new compute, network and storage resources to grow the cloud over time. The most crucial

part for cloud scalability is the amount of time required to set up and run the cloud and then scale up, as well as minimising the operational costs. For all this, what is needed is an automated deployment and configuration infrastructure that incorporates configuration management systems.

An automated deployment system installs and configures the operating system on new servers, without intervention, after the absolute minimum amount of manual work, including physical racking, MAC-to-IP assignment and power configuration.

OpenStack deployment can be done using any one of the following three important models:

- Enterprise based premise distribution
- Private cloud deployment
- OpenStack as a Service

In order to manage OpenStack cloud infrastructure, the systems administrators have to manage the configuration and the orchestration of the cloud services. Many open source tools are available to install, manage

and run the OpenStack cloud. The most important question is how to choose a deployment tool. This article lists many open source tools available for OpenStack deployment.

Chef

Chef is regarded as a powerful automation framework, which enables systems administrators to deploy servers and applications to any physical, virtual or cloud location, regardless of infrastructure size. Chef automates the process of managing configurations, making sure that systems are configured correctly and consistently. All the updates are applied dynamically, so that any changes on a running environment or hardware can be made without any code changes throughout the production.

Chef includes cookbooks for working with different options of OpenStack like Keystone. It makes use of Ruby as the scripting language, and includes a searchable portal where one can find community-contributed recipes and cookbooks. It has a number